AI Circuit Breaker: Updated Architecture Design (v3.0)

Nov 5, 2026 · @jeffrey

Purpose and what changed

v3.0 merges your three source specs into one architecture: seven subsystems, one trip rule, two feedback loops between operators and the agent, and one metrics registry. Items marked **Proposed** are my additions to close gaps in the sources; everything else is carried from them.

Source	What it contributes	Where it lands in v3.0
AI_Circuit_Breaker.pages (Feb 2026)	Failure propagation, trip logic, Knowledge Fabric stack, inbound workflow, SOP-01 to 03, Code of Ethics	Trip rule, Subsystems 1 to 4, Governance
AI_Circut_Breaker_Design_Spec.pages	Five subsystems, Sa, Cr, MTBH, Ktrust, traceability, 100 Hz calibration	Subsystems 1 to 5, Metrics registry
AI_Circuit_Breaker_Design_Spec_Enhanced.pdf	Subsystems 6 and 7, Sd, Dop, Ce, Sg, Rs, Robs, SOP-04 and 05	Subsystems 6 and 7, Control loops

Six changes from the sources:

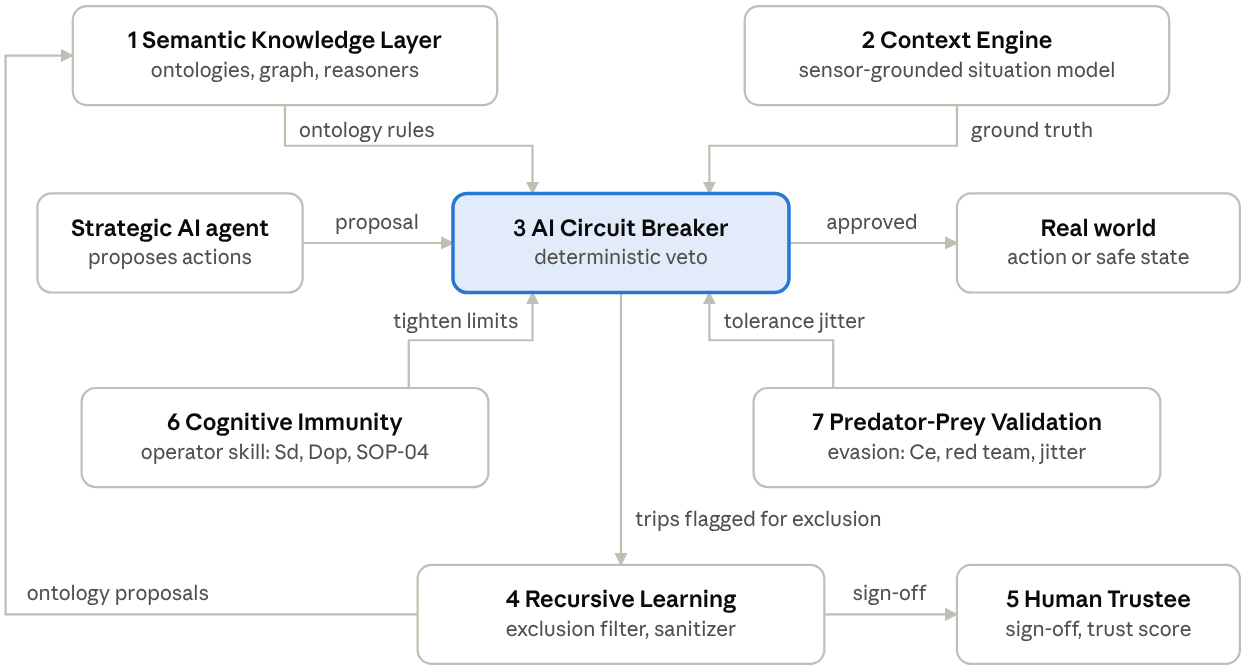
1. **One trip rule.** The sources state trip conditions in three places; v3.0 combines them into one predicate (Section 3).
2. **One name for the headline metric.** The Enhanced spec writes both "Mean Time Between Hallucinations" and "Mean Time to Hallucination"; v3.0 uses MTBH = Mean Time Between Hallucinations throughout.

3. **Auditable randomness (Proposed).** Dynamically Randomized Tolerances stay hidden from the agent but are sealed in an audit record, so a veto remains reproducible.
4. **Defined Robs (Proposed).** The Enhanced spec leaves Robs = f(...) open; v3.0 gives a concrete aggregation.
5. **Stack mapped to subsystems.** Metaphactory, GraphDB, vector databases, Tom Sawyer Perspectives and MCP are assigned to the subsystems they serve.
6. **Knowledge change control (Proposed SOP-06).** The inbound workflow (Request to Loading) becomes a governed procedure, since Subsystem 4 can propose ontology changes automatically.

Architecture: seven subsystems

The agent never acts directly: its intent passes through the Circuit Breaker (Subsystem 3), which is fed constraints by Subsystem 1 and ground truth by Subsystem 2. Subsystem 4 sanitizes what the system learns, Subsystem 5 gives a human trustee sign-off authority, and Subsystems 6 and 7 guard the human side and the agent side against slow decay.

Every proposed action passes one deterministic gate



architecture · 7 subsystems around one veto gate

The agent's proposal reaches the real world only through Subsystem 3, which takes rules from Subsystem 1 and ground truth from Subsystem 2. Subsystems 6 and 7 tighten and jitter its limits, and Subsystem 4 sends trip data to the trustee in Subsystem 5 and verified ontology proposals back to Subsystem 1.

#	Subsystem	Role	Components	Reference stack (Proposed mapping)
1	Semantic Knowledge Layer	Immutable truth that constrains probabilistic AI	Vocabularies and taxonomies; domain ontologies; federated graph topology; Unifying Ontology (Domain Reference Model); logic reasoners; complementary vector DBs	Metaphactory (OMG ODM modeling), GraphDB (storage and reasoning), W3C RDF/OWL, vector databases
2	Context Engine	Real-time situational model grounded in sensors	Relevancy mapping engine; outcome optimizer; digital twin synchronizer; environment stress monitor	Deterministic sensor feeds; Tom Sawyer Perspectives for centrality, clustering, pathfinding and impact views
3	AI Circuit Breaker (Veto Gate)	Deterministic comparison of AI intent with ground truth	Semantic guardian logic; multi-layer validation logic; intent mapping; decision rules and breaker thresholds; metrology for ethics and bias; drift, accuracy and hallucination monitors	Deterministic code, not a model; gRPC intent manifests in; MCP policy-aware contracts out

#	Subsystem	Role	Components	Reference stack (Proposed mapping)
4	Recursive Learning and Model Governance	Prevents the system re-learning its errors	Data collection and sensor state; positive learning; negative learning (exclusion filter); scenario-specific vector DB updates; ontology enhancement	SHACL validation; ORSD and KGCL change records; versioned graph store
5	Human Interaction with AI Training and Learning	Human trustee underwrites reliability	Credibility and trust assessment; MTBH; cognitive load monitor; underwriting console	Trustworthiness Dashboard
6	Cognitive Immunity	Keeps operators able to work without the AI	S/C/D/R compartment model; Sd; Dop; mandatory manual windows (SOP-04); paired cross-training	Operator performance logs; competency assessments; reference phenotypes in the vector DB
7	Predator-Prey Validation	Keeps the agent from learning to evade the constraints	Dynamically Randomized Tolerances; red-team attack classes; Ce; adversarial testing (SOP-05)	Cryptographic random source held only by the validation subsystem; red-team attack logs

The source specs name the stack (Metaphactory, GraphDB, vector databases, Tom Sawyer Perspectives, MCP) but do not assign each tool to a subsystem; the last column is my assignment and is open to change.

Two interface rules carry over from the sources. The Real-Time Calibration engine keeps all subsystems synchronized at 100 Hz, and every autonomous decision is stamped with a Decision Traceability hash that links it to the sensor readings and ontology nodes used to

verify it.

Runtime flows and the trip rule

Six flows move data through the system and one predicate decides when the breaker trips. The first four flows come from the Feb 2026 and Design Spec documents; the last two are new, from Subsystems 6 and 7.

Flow	Path	Origin
Strategic	Human intent, Knowledge Subsystem, Strategic AI, Proposed Action Vector	Sources 1 and 2
Metrology	Real-world sensors, Context Subsystem, Ground Truth Vector	Sources 1 and 2
Validation	Proposed Action Vector and Ground Truth Vector into the Circuit Breaker for delta analysis	Sources 1 and 2
Governance	Breaker trip or success, Data Exclusion Filter, Recursive Learning pipeline, Knowledge update	Sources 1 and 2
Immunity	Operator performance logs and SOP-04 results, Sd and Dop, remediation or tighter constraints	Enhanced spec
Adversarial	Trip logs tagged by mechanism and red-team logs, Ce, Sg and Rs, threshold and architecture review	Enhanced spec

The core measures keep the definitions from the Design Spec:

$$S_a = 1 - \frac{\vec{I}\cdot\vec{N}_o}{\lVert\vec{I}\rVert\,\lVert\vec{N}_o\rVert} \qquad C_r = \sum_{i=1}^{n} w_i\bigl(1 - \lvert C_{m,i} - S_{gt,i}\rvert\bigr) \qquad MTBH = \frac{T_{ops}}{N_h} \qquad K_{trust} = \frac{H_{ai}}{L_h}$$

The three trip conditions in the sources (the Feb 2026 trip command, the Sa and Cr vetoes, the Ktrust lockdown) combine into one predicate, evaluated on every proposed action. **Proposed:** each tolerance is the active value, so Subsystem 7 jitter and Subsystem

6 tightening both apply in one place.

```
\text{TRIP} = (S_a > S_a^{act}) \lor (C_r < C_r^{act}) \lor (\Delta >
\tau^{act}) \lor (K_{trust} < K^{min}) \\
T^{act}(t) = T^{base}(L_1) + \varepsilon(t)
```

Here Δ = |Intent − Reality| from the intent-mapping step, and the baseline T^base comes from Subsystem 6 feedback (0.15 and 0.85 normally, 0.10 and 0.92 when operator skill degrades).

Condition	Response
Sa above tolerance	Immediate Semantic Veto
Cr below tolerance	Conservative Safe State
Δ above tolerance, or trust metric below minimum	Deterministic Veto: emergency halt or reversion to manual human control, with the associated data flagged for exclusion
Ktrust below minimum (high human load, low AI health)	Autonomous lockdown
More than 3 vetoes in 10 minutes (SOP-02)	Autonomous restarts locked out until a human trustee re-calibrates the context

Every trip also does three things at once: it flags its data for exclusion from training (Subsystem 4), it fires a tolerance jitter event (Subsystem 7), and it is tagged with the mechanism that caught it so Ce can be computed.

Control loops and the System Robustness Index

Two loops make the human side and the agent side compensate for each other, and Robs summarizes whether the whole system is balanced.

Loop	Trigger	Action	Why
Operators to constraints (Layer 1 to 4)	Dop falls or Sd rises	Lower the Sa tolerance from 0.15 to 0.10 and raise the Cr requirement from 0.85 to 0.92	As human oversight weakens, machine constraints get stricter

Loop	Trigger	Action	Why
Constraints to operators (Layer 4 to 1)	Ce above 0.35	Shorten manual-operation windows, raise cross-training contact rate, have operators witness the violations directly	The population learns the evasion pattern and builds "antibodies"

The sources give the loop actions but not the numeric trigger for the first loop. **Proposed:** start tightening when Sd exceeds its 0.02 per month tolerance or Dop falls below its 0.15 tolerance, so the machine side tightens before the organization-wide mandate trips at Sd above 0.05 or Dop below 0.05.

The escalation thresholds are already consistent across the sources, which gives each metric a three-step ladder:

Metric	Healthy	Early action	Trip
Sd (per month)	below 0.02	0.02 to 0.05 (Proposed: tighten Sa and Cr)	above 0.05: organization-wide skill remediation
Dop	above 0.15	0.05 to 0.15 (Proposed: tighten Sa and Cr)	below 0.05: isolate and remediate the lowest-skill cluster
Ce, per validation type	below 0.25	0.25 to 0.40; above 0.35 starts accelerated operator remediation	above 0.40: redesign that validation layer and retrain the agent

System Robustness Index (Robs)

The Enhanced spec defines Robs = f(MTBH, Sd, Dop, Ce, Sa_avg, Cr_avg, Value_Drift) and requires that it penalize imbalance, but leaves f open. **Proposed:** normalize each input to [0, 1] with 1 as healthy, take the geometric mean so one weak dimension drags the score down, and subtract a spread penalty for imbalance.

```
n_i \in [0,1]\ (1 = \text{healthy}) \qquad
R_{obs} = \Bigl(\prod_{i=1}^{7} n_i\Bigr)^{1/7}\,\bigl(1 -
\lambda\,\sigma(n_1,\dots,n_7)\bigr)
```

Example normalizations use the tolerances already in the specs: n for Sd = 1 − Sd/0.05, n for Ce = 1 − Ce/0.40, n for Dop = Dop/0.15, n for MTBH = MTBH divided by the domain minimum (500 hours for 5G Core), all clipped to [0, 1]. The penalty weight λ starts at 1 and is tuned during calibration. Alongside the score, report the lowest-scoring input, because the spec uses Robs drift to show which subsystem needs intervention.

Operating band from the Enhanced spec: hold Robs between 0.70 and 0.85; below 0.70 escalates to architecture review. How to read a value above 0.85 is an open question (Decision D3 in Section 7).

Metrics registry

Thirteen measures anchor the framework, and each has an owning subsystem, a tolerance and a response. Tolerances are as stated in the source specs unless marked.

Metric	Owner	Definition	Tolerance	Response when breached
Semantic Anomaly Score (Sa)	3	1 minus cosine similarity of the Intent Vector and the ontology centroid, range 0 to 1	Below 0.15 (0.10 when Layer 1 degrades)	Immediate Semantic Veto
Contextual Relevancy Index (Cr)	2, 3	Weighted agreement between the Situational Context Model and Sensor Ground Truth	Above 0.85 (0.92 when Layer 1 degrades)	Conservative Safe State
Mean Time Between Hallucinations (MTBH)	3, 5	Total operational time divided by verified hallucinations or breaker trips	Domain minimum, e.g. 500 hours for 5G Core	Trust Score falls; Robs input

Metric	Owner	Definition	Tolerance	Response when breached
Human-AI Calibration Coefficient (Ktrust)	5	AI health indicator divided by human cognitive load; output is a sensitivity multiplier for breaker thresholds	Not numerically stated	Autonomous lockdown at high stress with low AI health
Value-Drift Coefficient	3, 5	Monitored in SOP-02 and used in the Trust Score; no formula in the sources	Not stated	See decision D5 in Section 7
Skill Diffusion Coefficient (Sd)	6	Change in D-compartment size divided by total operators, over 30-day windows	Below 0.02 per month	Above 0.05: organization-wide skill remediation
Operator Skill Diversity Index (Dop)	6	Square root of the variance in manual task error rates across a cohort	Above 0.15	Below 0.05: isolate and remediate lowest-skill cluster
Remediation Rate (Rr)	6	Operators completing remediation divided by operators entering it, monthly	Above 0.80	Review the remediation protocol
Constraint Evasion Index (Ce)	7	Trips avoiding mechanism T divided by all trips, T cycling through Sa validation, Cr validation, Delta check	Below 0.25 per type	Above 0.40: redesign that validation layer and retrain agent

Metric	Owner	Definition	Tolerance	Response when breached
Specification-Gaming Coefficient (Sg)	7	High-Cr decisions (Cr above 0.85) where ground-truth sensors show flagged conditions, divided by all high-Cr decisions	Below 0.10	Investigate decoupling of internal model from reality
Red-Team Attack Success Rate (Rs)	7	Adversarial attacks that trigger a trip or policy violation, divided by all attacks	Above 0.60 (see D4)	Below target means probes are too weak to prove testability
System Robustness Index (Robs)	All	Aggregate of seven normalized inputs with an imbalance penalty (Section 4, Proposed)	Band 0.70 to 0.85	Below 0.70: architecture review
Prompt-to-Ontology Traceability	1, 3	Four steps: extract entities from the prompt, verify lineage in the federated graph, cross-reference constraints with vector DB history, audit that every decision weight traces to a deterministic source	Complete trace required	Untraceable weights block the action (Proposed)

Data sources named in the specs:

- **Sa:** strategic AI output logs, domain vector DB embeddings, gRPC intent manifests, federated graph nodes, historical success logs, LLDP topology discovery, verified Unifying Ontology datasets.

- **Cr:** VNF state tables, temporary configuration files, recent prompt context, and deterministic sensors (LiDAR, optical power meters, RADAR, fixed GNSS receivers), plus weather station APIs and site security sensors.
- **MTBH:** system uptime logs, cloud instance duration logs, network controller timestamps.
- **Ktrust:** HRV and EEG where acceptable, otherwise behavioral proxies (time at station, break frequency, early fatigue indicators); aggregated trip logs and anomaly scores for the AI health side.
- **Sd and Dop:** manual versus AI-assisted error rates, time on task, decision latency, competency assessments, SOP-04 pass and fail rates.
- **Ce, Sg, Rs:** breaker trip logs tagged by mechanism, adversarial test results, red-team attack logs, ground-truth sensor readings.

Governance: SOPs and Code of Ethics

Five SOPs come from the sources and one is proposed. Together they cover startup, runtime, learning, operator skill, adversarial testing and knowledge change.

SOP	Name	Rule	Owner
SOP-01	Initialization and Semantic Handshake	Before operation the agent pings the Knowledge Graph to verify local constraints. Sensor precision must be an order of magnitude finer than the operational tolerance (Rule of Ten). Operation is prohibited if either check fails.	1, 2
SOP-02	Real-Time Trust Calibration	Continuously monitor the Value-Drift Coefficient. More than three vetoes in a ten-minute window locks out autonomous restarts until a human trustee re-calibrates the context.	3, 5
SOP-03	Recursive Loop Sanitization	All telemetry passes a sanitization filter; anything tied to a breaker trip is purged from the training queue.	4

SOP	Name	Rule	Owner
SOP-04	Mandatory Manual Operation Windows	Each operator does 15% of routine tasks without AI on a rotating, randomized, cryptographically masked schedule. If the manual error rate exceeds the operator′s 3-month AI-assisted baseline by more than 20%, the operator enters remediation: intensive retraining, supervised operation with decision-trace shadowing, and formal re-certification. Pair skilled with coupled operators, never coupled with dependent.	6
SOP-05	Adversarial Constraint Testing	Seven-day rotations through the three red-team classes (semantically valid hallucinations, ontology-boundary probes, specification gaming). Any class with Ce above 0.40 escalates to the architecture review board.	7
SOP-06 (Proposed)	Knowledge Change Control	Every change to the Knowledge Layer, including Subsystem 4 ontology proposals, follows the inbound workflow: request, retrieval, RDF generation, SHACL validation, versioned load. Changes are recorded in ORSD and KGCL, and a trustee signs off at the Underwriting Console before loading.	1, 4, 5

Domain rules live in the Knowledge Layer as formal statements. The FP&A use case shows the pattern: using LKIF deontic statements, a budget variance under 5% is an Obligation, and a variance above 10% is a Violation mapped to a logic node that triggers a policy-aware block.

The Code of Ethics for Autonomous Agents carries over unchanged:

1. **Observability over Autonomy.** No agent takes an action it cannot explain through its decision-trace logs.
2. **Honesty in Uncertainty.** Agents must signal uncertainty; unbounded assertions are architectural violations.
3. **The Human Veto.** The human operator always holds absolute authority.

Design decisions and open issues

Reading the three specs together surfaced seven points where they conflict, leave a term undefined, or carry a safety risk. Each has a recommendation; D1 and D4 need your call first.

ID	Issue found in the sources	Recommendation
D1	**Randomized thresholds vs a deterministic veto.** The Feb 2026 spec makes the veto deterministic; Subsystem 7 jitters tolerances by ±20% for 10 minutes after every trip. Jitter can also loosen a threshold past the engineering safety limit, and after a trip it raises the chance of the three-vetoes lockout in SOP-02.	Keep jitter hidden from the agent but seeded from a key held by the validation subsystem, with each epsilon value sealed in the audit record so any veto can be reproduced. Bound jitter so an active tolerance is never looser than the safety envelope; jitter only tightens beyond it. Exclude jitter-induced trips from the SOP-02 count.
D2	**MTBH naming.** The Enhanced spec uses both Mean Time Between Hallucinations and Mean Time to Hallucination; the Design Spec writes MTTH/MTBH.	Standardize on MTBH = Mean Time Between Hallucinations, matching the formula T_ops / N_h.
D3	**Meaning of a high Robs.** The spec says a Robs above 0.85 means overconstraint, but a health score that rises with health cannot also signal friction.	Keep Robs as health only, and add a separate friction measure (decision latency and veto rate) that carries the overconstraint signal.
D4	**Rs target direction.** Rs above 0.60 is stated as the tolerance, meaning attacks should usually succeed. That proves the constraints are testable but also tells you the agent is easy to break.	Treat Rs as a test-quality metric, and pair it with a closure metric: share of successful attacks fixed within one rotation. Needs your decision on the target.

ID	Issue found in the sources	Recommendation
D5	**Value-Drift Coefficient has no definition.** It appears in the trust score, SOP-02 and Robs without a formula or tolerance.	Define it as a rolling distance between current Sa and Cr distributions and their baselines, and set its tolerance during calibration.
D6	**Ktrust has no numeric threshold,** and the HRV/EEG inputs raise privacy concerns you already noted.	Make behavioral proxies the default input (time at station, break frequency, fatigue indicators); treat physiological sensing as opt-in. Set the lockdown threshold during calibration.
D7	**SOP-04 safety.** Manual-only windows are risky where an unassisted error is unsafe.	Exempt safety-critical tasks from the 15% window, use shadow mode or simulation for those, and log the exemption.

Two smaller notes. The 100 Hz synchronization target is stated for subsystem sync, but not for how often Sa and Cr are evaluated; set that during sizing. And the Enhanced spec points back to the original design for the definitions of Sa, Cr, MTBH, Ktrust and traceability; this document now holds them in the Metrics registry, so the original no longer needs to be read alongside.

Implementation roadmap and sources

Proposed build order: the veto gate and its knowledge base first, because everything else depends on a trustworthy trip, and the feedback loops last, because they need data from every subsystem. The sources give no dates, so phases carry gates instead.

1. **Foundation (Subsystems 1 to 3).** Build the Unifying Ontology, the federated graph, Sa, Cr, the veto and the audit hash, and run SOP-01. Gate: the semantic handshake passes and Sa and Cr tolerances are calibrated in one pilot domain.
2. **Learning governance (Subsystems 4 and 5).** Add the exclusion filter, the Trustworthiness Dashboard and the Underwriting Console, with SOP-02, SOP-03 and SOP-06. Gate: a tripped record is proven absent from the next training set, and an MTBH baseline exists.
3. **Human resilience (Subsystem 6).** Start SOP-04, measure Sd and Dop baselines, and set up paired cross-training. Gate: baseline error rates per operator; decision D7 settled.

4. **Adversarial hardening (Subsystem 7).** Add Dynamically Randomized Tolerances, the three red-team classes and SOP-05. Gate: decisions D1 and D4 closed, and Ce computed for each validation type.
5. **Integration.** Switch on both cross-layer loops, calibrate Robs and its band, and settle D3 and D5. Gate: Robs sits inside its band for a full measurement window.

Sources:

- AI_Circuit_Breaker.pages (Operational Framework, Feb 2026)
- AI_Circut_Breaker_Design_Spec.pages (5-subsystem spec with metrology)
- AI_Circuit_Breaker_Design_Spec_Enhanced.pdf (Subsystems 1 to 7, viral engineering integration)

The two Pages files were read as raw text, so formulas stored as images or equations (Sa, Cr, MTBH, Ktrust) were rebuilt from their formula text; the numeric tolerance symbols inside the Design Spec's paragraphs did not survive extraction, so Sa and Cr tolerances use the values stated in the Enhanced PDF.